SEQUENCE 3

Straight lines in the plane

1–16 October 2026

3.1 Direction vector and Cartesian equation

Key point · Definition. A direction vector of a line is a **nonzero** vector having the direction of that line. If $A \neq B$, $\overrightarrow{AB}$ directs (AB). Every line has an equation $ax + by + c = 0$ with $(a, b) \neq (0, 0)$. A direction vector is $\begin{pmatrix} -b \\ a \end{pmatrix}$.

Example 1. For $A(-2, 4)$ and $B(6, 2)$, calculate $\overrightarrow{AB}$ then check that $\vec{u}(4, -1)$ directs (AB).

Solution. $\overrightarrow{AB} = (8, -2) = 2\vec{u}$. The vector $\vec{u}$ is nonzero and collinear with $\overrightarrow{AB}$; it therefore directs (AB).

Key point · Establishing an equation with the determinant. The line through $A(x_A, y_A)$ with direction vector $\vec{u}(\alpha, \beta) \neq \vec{0}$ has equation

$$\beta(x - x_A) - \alpha(y - y_A) = 0.$$

Proof to write.

Solution. A point $M(x, y)$ belongs to the line if and only if $\overrightarrow{AM}$ is collinear with $\vec{u}$, including for $M = A$. Now

$$\det(\overrightarrow{AM}, \vec{u}) = (x - x_A)\beta - (y - y_A)\alpha.$$

Setting the determinant to zero gives the stated equation. Expanding, $a = \beta$, $b = -\alpha$ and $c = -\beta x_A + \alpha y_A$; since $\vec{u} \neq \vec{0}$, $(a, b) \neq (0, 0)$.

Watch out. The zero vector directs no line. The sign of the determinant depends on the order of the vectors, but whether it vanishes does not.

3.2 Determining the equation of a line

Method. From a point and a direction vector: choose a, b then replace x, y by the coordinates of the point to find c. From two distinct points: first calculate a direction vector.

Example 2. Determine a Cartesian equation of the line through $A(2,-1)$ with direction vector $\vec{u}(3,2)$.

Solution. $a = 2$, $b = -3$ will do. The line therefore has an equation $2x - 3y + c = 0$. Since A belongs to it, $4 + 3 + c = 0$, so $c = -7$. An equation is $2x - 3y - 7 = 0$. Check: $2 \times 2 - 3 \times (-1) - 7 = 0$ and $(3,2)$ is indeed a direction vector.

Example 3. Determine a Cartesian equation of (AB) for $A(3,3)$ and $B(2,-2)$.

Solution. $\overrightarrow{AB} = (-1,-5)$ directs the line. We can choose $(a,b) = (5,-1)$. With A, $15 - 3 + c = 0$, so $c = -12$. Thus $5x - y - 12 = 0$. For B, $10 + 2 - 12 = 0$, which confirms the result.

Example 4. Does the point $P(1,2)$ belong to $d : 2x - 3y + 4 = 0$? Does the point $Q(2,1)$ belong to it?

Solution. For P, $2 - 6 + 4 = 0$: yes. For Q, $4 - 3 + 4 = 5 \neq 0$: no.

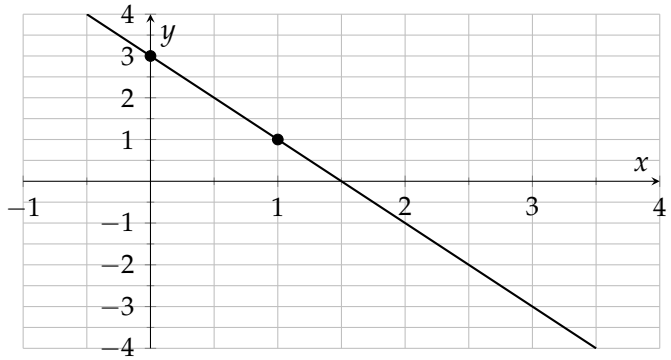
3.3 Slope-intercept equation, slope and affine function

Key point · Definitions. A non-vertical line has a unique slope-intercept equation $y = mx + p$. m is the slope, p the y-intercept. It represents the affine function $x \mapsto mx + p$; $(1, m)$ is a direction vector. A vertical line has an equation $x = k$; it is not the graph of a function of x. A horizontal line has an equation $y = k$.

Key point · Calculating a slope. If $x_A \neq x_B$, the slope of (AB) is $m = \dfrac{y_B - y_A}{x_B - x_A}$. To find p, we then use $y_A = mx_A + p$.

Example 5. Determine the slope-intercept equation of $2x + 3y - 1 = 0$, then draw $d : y = -2x + 3$.

Solution. $3y = -2x + 1$, so $y = -\frac{2}{3}x + \frac{1}{3}$. For d, plot $(0,3)$ and $(1,1)$, then draw the line; a move of 1 along the x-axis corresponds to -2 on the y-axis.



Example 6. Find the affine function f such that $f(0) = 3$ and $f(-1) = 5$, then the affine function g whose graph passes through $C(-1, 2)$ and $E(5, -2)$.

Solution. $f(x) = -2x + 3$, since $m = (3-5)/(0+1) = -2$. For g, $m = (-2-2)/(5+1) = -2/3$ and $p = 2 - 2/3 = 4/3$, so $g(x) = -\frac{2}{3}x + \frac{4}{3}$.

3.4 Relative positions of two lines

Key point · Properties. Two lines are either intersecting, strictly parallel or coincident. For $y = mx + p$ and $y = m'x + p'$:

- if $m \neq m'$, they intersect;
- if $m = m'$ and $p \neq p'$, they are strictly parallel;
- if $m = m'$ and $p = p'$, they coincide.

For Cartesian equations, we compare the directions with the determinant. If the directions are collinear, we then test the membership of a point to distinguish coincident lines from strictly parallel ones.

Example 7. Determine an equation of d through $A(2,3)$ and parallel to $d' : x - 3y - 3 = 0$.

Solution. d' has direction vector $(3,1)$. We look for $d : x - 3y + c = 0$. Since $A \in d$, $2 - 9 + c = 0$, so $c = 7$. Thus $d : x - 3y + 7 = 0$. The two lines are strictly parallel, because the constants differ for the same coefficients of x, y.

Example 8. Study the relative positions:

1. $2x - y + 1 = 0$ and $4x - 2y + 2 = 0$;
2. $x = 2$ and $x = -1$;
3. $y = 2x - 3$ and $y = -x + 3$.

Solution. 1. The second equation is twice the first: coincident lines. 2. Two distinct vertical lines: strictly parallel. 3. The slopes 2 and -1 differ: intersecting. Their intersection satisfies $2x - 3 = -x + 3$, so $x = 2$ and $y = 1$.

3.5 Intersection: solving by substitution

Key point · Definition. Solving a system of two equations in x, y means finding all the pairs (x, y) satisfying both equations simultaneously. For two equations of lines: an intersection gives one solution pair; strictly parallel lines give no solution; coincident lines give infinitely many solutions.

Method · Substitution. Isolate one unknown in one equation, substitute its expression into the other, solve, then find the second unknown. Check the pair in both original equations.

Example 9. Solve $\begin{cases} -2x + y + 3 = 0, \\ 3x - 5y - 2 = 0. \end{cases}$

Solution. The first equation gives $y = 2x - 3$. Substituting into the second: $3x - 5(2x - 3) - 2 = 0$, that is $-7x + 13 = 0$. Thus $x = 13/7$ then $y = 5/7$. The solution set is $S = \{(13/7, 5/7)\}$. The substitutions give $(-26 + 5 + 21)/7 = 0$ and $(39 - 25 - 14)/7 = 0$.

Example 10. Solve by substitution:

$$\text{(a)} \begin{cases} -2x + y + 3 = 0 \\ x - 4y + 5 = 0 \end{cases} \qquad \text{(b)} \begin{cases} 2x + 3y - 10 = 0 \\ 4x + 5y - 16 = 0. \end{cases}$$

Solution. (a) $y = 2x - 3$ then $x - 4(2x - 3) + 5 = 0$, so $x = 17/7$, $y = 13/7$.
(b) $x = (10 - 3y)/2$; $2(10 - 3y) + 5y - 16 = 0$, so $y = 4$, then $x = -1$.

3.6 Linear combination and Python

Method · Linear combination. Multiply the equations by nonzero numbers to obtain opposite coefficients for one unknown; add them to eliminate that unknown. Keep one equation of the system to calculate the other unknown.

Example 11. Solve $\begin{cases} 2x+3y+1=0 \\ 3x-4y-2=0. \end{cases}$ Then redo the two systems of the previous example by linear combination.

Solution. Multiplying the first equation by 3 and the second by -2 gives $6x+9y+3=0$ and $-6x+8y+4=0$. The sum gives $17y+7=0$, so $y=-7/17$. Then $2x-21/17+1=0$, so $x=2/17$. For (a), add four times the first equation to the second: $-7x+17=0$, so $(x,y)=(17/7,13/7)$. For (b), subtract twice the first from the second: $-y+4=0$, so $(x,y)=(-1,4)$.

```
def determinant(u, v):
    return u[0]*v[1] - u[1]*v[0]

def alignes(A, B, C):
    AB = (B[0]-A[0], B[1]-A[1])
    AC = (C[0]-A[0], C[1]-A[1])
    return determinant(AB, AC) == 0
```

Example 12. Predict `alignes((0,0), (2,1), (6,3))` then `alignes((0,0), (2,1), (6,4))`.

Solution. `True`, then `False`: the determinants are 0 and 2. The exact test is suitable here for integer coordinates; floating-point approximations require taking numerical errors into account.